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PROJECT PROPOSAL

Project Title:

Ransomware Simulation & Backup Recovery System

1. Problem Statement:

Ransomware is a serious cybersecurity threat that encrypts user files and makes them inaccessible. After encryption, users cannot open their files and may lose important data. Many systems do not have proper protection or backup mechanisms to recover from such attacks. The purpose of this project is to simulate a ransomware file encryption attack in a controlled and safe environment. The project will show how files become unreadable after encryption and how a backup recovery mechanism can restore the original data. The main objective is to understand how ransomware attacks affect file security and to demonstrate how backup strategies can protect against permanent data loss.

2. Scope:

This project will:

- Simulate file encryption using a simple character-based encryption technique.
- Show how encrypted files become unreadable.
- Create a backup of the original file before encryption.
- Restore the original file using the backup mechanism.
- Demonstrate both the attack (encryption) and the defense (recovery).

3. Limitations:

- The encryption method will be simple and for educational purposes only.
- The project will only work on selected text files.
- It will not involve real malware, real hacking, or network attacks.
- The system will be implemented in a controlled academic environment only